



Mobile Forensics: Digital Forensic Tool for WhatsApp Data Analysis

(Whisper-WA)

ABSTRACT

Whisper-WA is a specialized desktop digital forensics platform designed to address critical gaps in **WhatsApp data analysis for cybercrime investigations in Saudi Arabia**. The system automates a forensically sound acquisition pipeline: ADB-based database extraction, Crypt14/15 decryption, SQLite3 parsing, Arabic-aware GUI analysis, and PDF forensic report generation with full chain-of-custody documentation.

80% of digital crimes in Saudi Arabia involve WhatsApp — making it the most critical source of forensic evidence. (Saudi Dept. of Criminal Evidence)

PROBLEM STATEMENT & RESEARCH GAP

Problem Statement

WhatsApp is the dominant communication platform in Saudi cybercrimes, implicated in fraud, extortion, blackmail, and identity theft. Investigators are forced to rely on screenshots as primary evidence — artefacts that lack metadata, are easily falsified, and hold no legal standing in formal proceedings. No existing tool provides a free, Arabic-native, forensically sound solution tailored to the Saudi investigative context.

Research Gap

Methodological

Existing tools require WhatsApp APK downgrading, modifying the target device and compromising evidence integrity

Incomplete Recovery

Many tools retrieve only thumbnails, failing to recover full media, documents, or voice messages

No Arabic Support

No tool provides native RTL rendering or Arabic-formatted forensic reporting

Cost Barrier

Commercial platforms cost thousands of dollars, inaccessible to academic institutions and regional agencies

No Saudi Context

No tool is tailored to Saudi Arabia's legal framework or cybercrime investigation requirements

METHODOLOGY

Development Approach — Adaptive Agile-Scrum The system was developed across six iterative sprints, each delivering a validated forensic subsystem before the next commenced, ensuring continuous alignment between implementation and investigative requirements.

Requirements Basis

- Research papers in WhatsApp and mobile forensics
- DFIR & SOC-oriented discussion
- Insights from Black Hat cybersecurity events
- ISO/IEC 27037 considerations

System Architecture — Three-Tier Model

Presentation

Arabic RTL GUI



Logic Layer

Python Engine



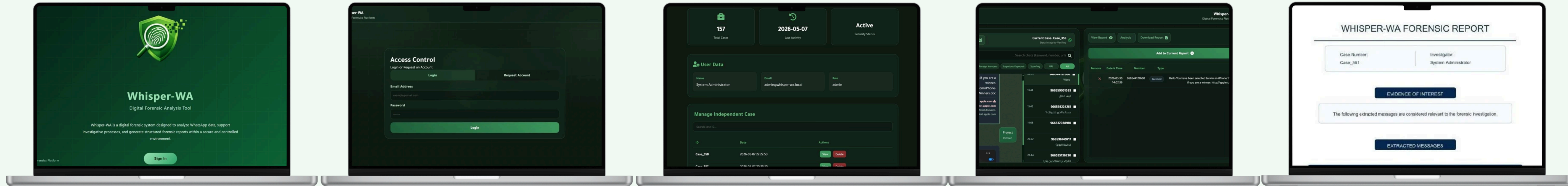
Data Layer

SQLite3

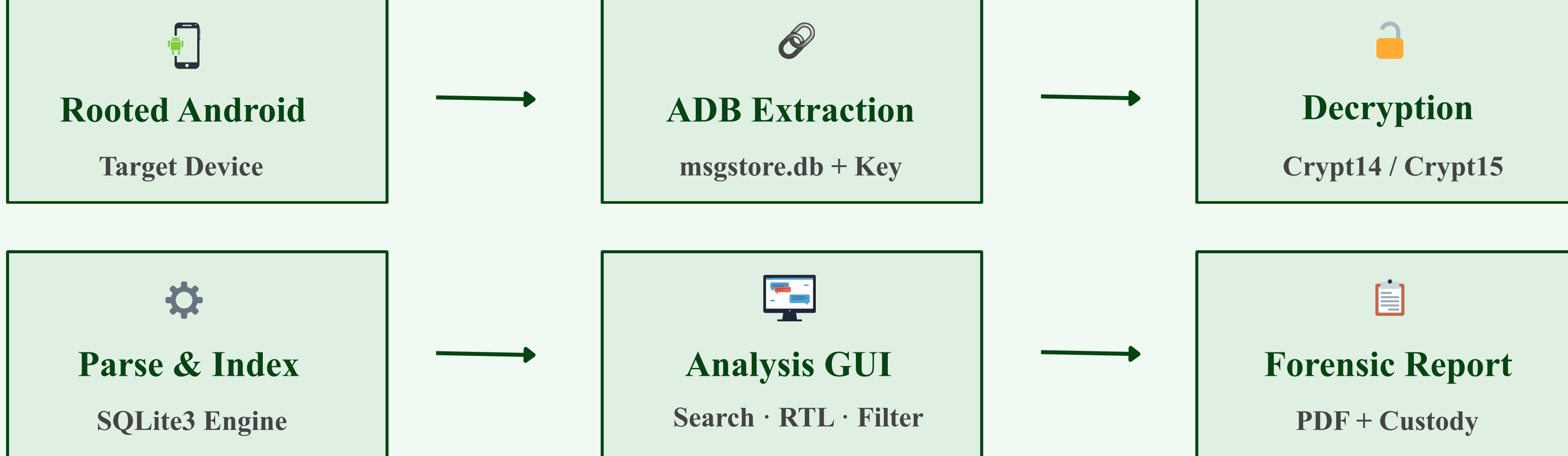
TECHNOLOGY STACK

 Python Core engine & forensic logic	 ADB Android device bridge & extraction	 SQLite3 Database parsing & indexing	 wa-crypt-tools Crypt14/15 decryption library
 HTML5 GUI structure & layout	 CSS3 Styling & Arabic RTL support	 JavaScript Interface interactivity	 Windows OS Desktop host platform

INTERFACES



SYSTEM ARCHITECTURE & WORKFLOW



KEY FEATURES

Automated Decryption

Crypt14/15 — no downgrade, evidence fully preserved

Native Arabic + RTL

Full RTL rendering & Arabic forensic reports — unique

Advanced Search & Filter

Date, sender, keyword, conversation thread

Hash Integrity (SHA-256)

Provably unmodified data from acquisition to report

Forensic PDF Reports

Court-admissible + chain-of-custody

Evidence Correlation

Reconstruct timelines accurately

COMPARATIVE ANALYSIS

Feature	Belkasoft	Cellebrite	Oxygen	WA DB	Whisper-WA
Free Access	✗	✗	✗	✓	✓
No Downgrade	✓	✓	✓	✗	✓
Arabic + RTL	~	~	~	~	✓ Naitve
Chain of Custody	~	✓	✓	✗	✓
Hash Integrity	~	✓	✓	✗	✓
Saudi Context	✗	✗	✗	✗	✓

EVALUATION

Whisper-WA was evaluated through a structured multi-level testing framework — unit testing across 5 core classes (25 cases), integration testing across 4 data flow points, functional verification of FR-01 through FR-06, and forensic soundness testing confirming read-only enforcement, SHA-256 hash integrity, and complete audit trail generation in alignment with ISO/IEC 27037

CONCLUSION & FUTURE WORK

Whisper-WA delivers a free, forensically sound, and Arabic-native platform that replaces unreliable screenshot evidence with legally admissible, integrity-verified digital forensics — automating the full WhatsApp acquisition pipeline from ADB extraction to court-ready PDF reporting, aligned with ISO/IEC 27037 and purpose-built for Saudi cybercrime investigation.

Futuer Work:

- Full multimedia recovery
- Non-rooted device support
- iOS platform extension
- Cloud backup acquisition
- Deleted message recovery via SQLite WAL files



Live Demo



GitHub